Laplace's equation

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

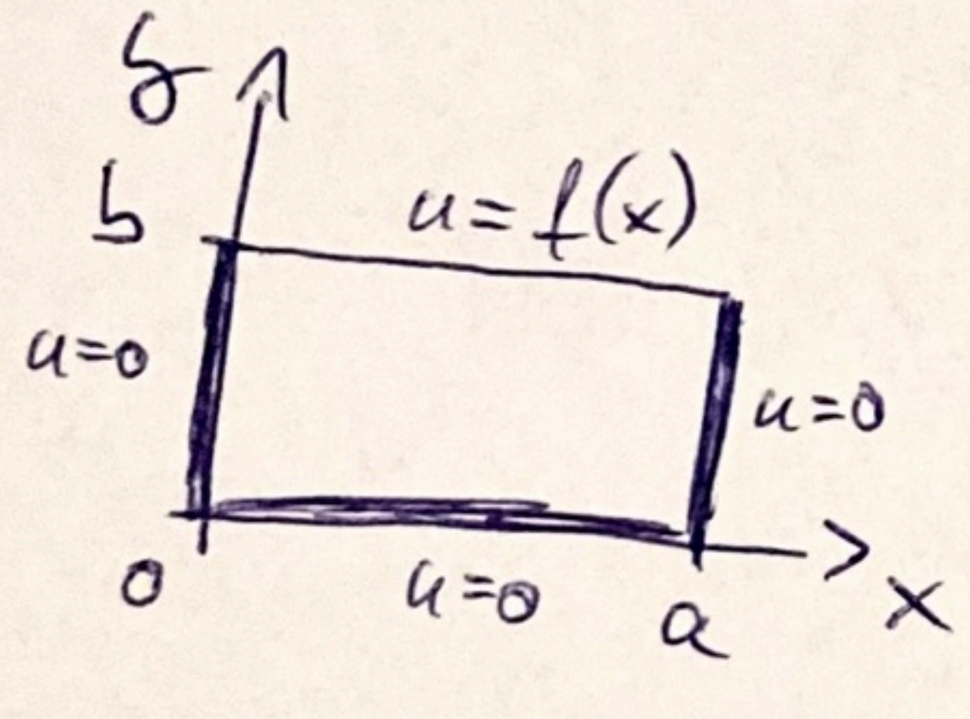
Separation of variables $\quad u(x,y) = F(x)\,G(y)$

$$\frac{1}{F}\frac{d^2F}{\partial x^2} = -\frac{1}{G}\frac{d^2G}{\partial y^2} = -k$$

$$F'' = -kF \qquad G'' = kG$$

Sign of k comes from boundar conditions

Example



$$u(0,y) = u(a,y) = u(x,0) = 0$$
$$u(x,b) = f(x)$$

$$F(0) = F(a) = 0$$

$$F_n(x) = \sin(\lambda_n x) \qquad \lambda_n = \frac{n\pi}{a}$$

$$\Rightarrow k_n = \lambda_n^2$$

$$G_n'' = \lambda_n^2 G_n$$

$$G_n(y) = A_n e^{\lambda_n y} + B_n e^{-\lambda_n y}$$

$$G_n(0) = 0 \quad \Rightarrow \quad B_n = -A_n$$
$$G_n(y) = A_n \sinh(\lambda_n y)$$

General solution

$$u(x,y) = \sum_{n=1}^{\infty} A_n \sin(\lambda_n x)\sinh(\lambda_n y)$$

Last boundary condition

$$f(x) = u(x,b) = \sum_{n=1}^{\infty} A_n \sin(\lambda_n x) \sinh(\lambda_n b)$$

Again Fourier series of $f(x)$

$$b_n = \frac{2}{a} \int_0^a f(x) \sin(\lambda_n x)\, dx = A_n \sinh(\lambda_n b)$$

Example $f(x) = u_b$ constant

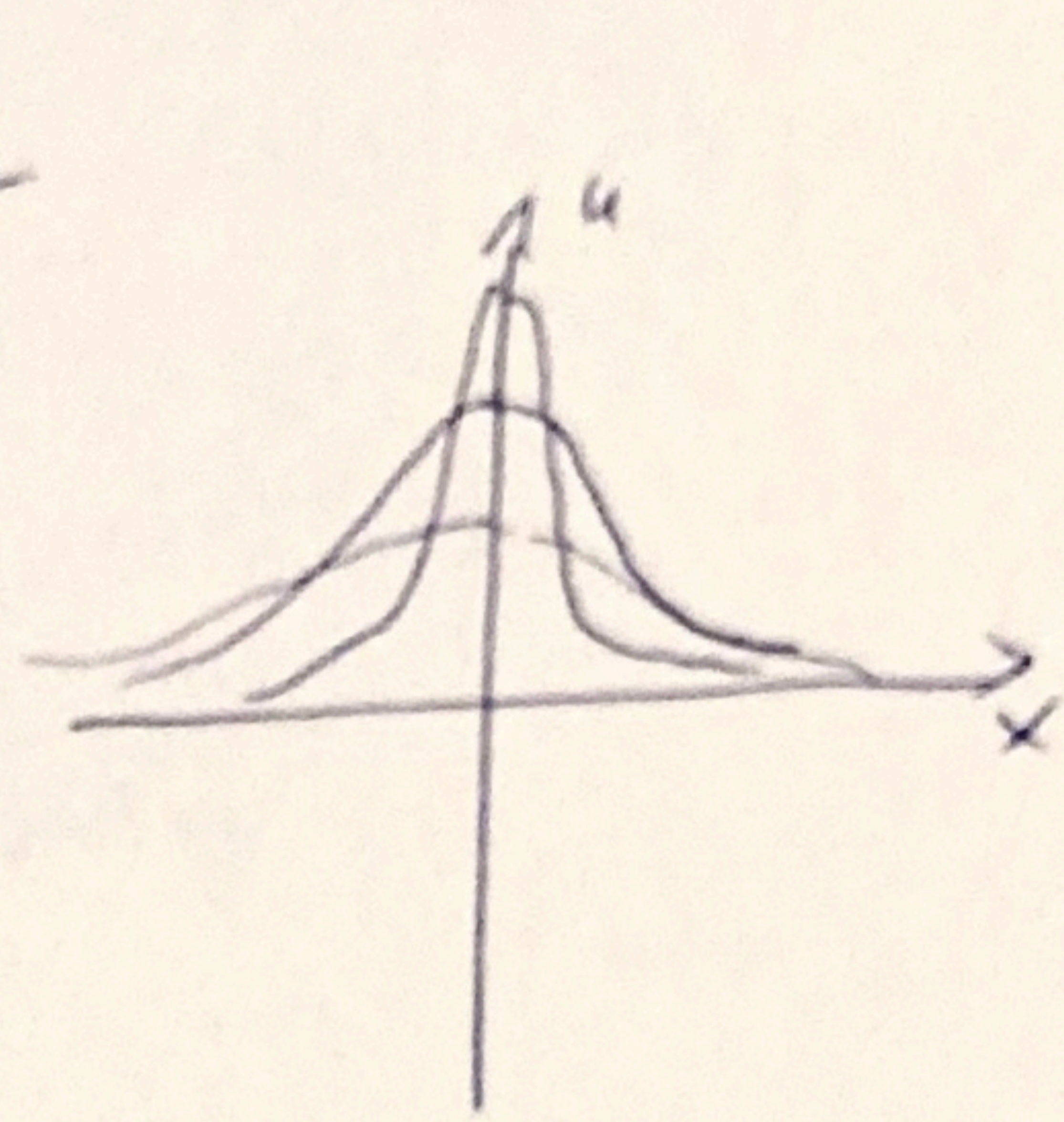
$$b_n = \frac{2}{a} \int_0^a u_b \sin\left(\frac{n\pi}{a} x\right) dx = -\frac{2u_b}{a} \frac{a}{n\pi} \left[\cos \frac{n\pi}{a} x\right]_0^a =$$

$$= -\frac{2u_b}{n\pi} \left((-1)^n - 1\right) = \begin{cases} 0 & \text{if } n \text{ even} \\ \frac{4u_b}{n\pi} & \text{if } n \text{ odd} \end{cases}$$

$$A_n = \frac{b_n}{\sinh(\lambda_n b)} = \begin{cases} 0 & \text{if } n \text{ even} \\ \frac{4u_b}{n\pi \sinh\left(\frac{n\pi b}{a}\right)} & \text{if } n \text{ odd} \end{cases}$$

Heat equation in infinite space

Spreading Gaussian is a solution



$$u_0(x,t) = \frac{1}{\sqrt{t}}\, e^{-\frac{x^2}{4c^2t}}$$

$$\frac{\partial u_0}{\partial t} = \left(-\frac{1}{2}\frac{1}{t^{3/2}} - \frac{1}{\sqrt{t}}\frac{x^2}{4c^2t^2}\right) e^{-\frac{x^2}{4c^2t}}$$

$$\frac{\partial u_0}{\partial x} = -\frac{1}{\sqrt{t}}\frac{x}{2c^2t}\, e^{-\frac{x^2}{4c^2t}}$$

$$\frac{\partial^2 u_0}{\partial x^2} = \left(-\frac{1}{\sqrt{t}\cdot 2c^2t} + \frac{1}{\sqrt{t}}\frac{x^2}{4c^4t^2}\right) e^{-\frac{x^2}{4c^2t}}$$

$$\frac{\partial u_0}{\partial t} = c^2 \frac{\partial^2 u_0}{\partial x^2} \quad \checkmark$$

It is also a solution shifted

$$u_v(x,t) = u_0(x-v,t)$$

Linear combinations are also solutions

$$u'(x,t) = \int_{-\infty}^{\infty} f(v)\, u_0(x-v,t)\, dv$$

For $t \to 0$ it is Dirac delta

$$\lim_{t\to 0} u_0(x,t) = 2c\sqrt{\pi}\,\delta(x)$$

$$\lim_{t\to 0} u'(x,t) = \int_{-\infty}^{\infty} f(v)\, 2c\sqrt{\pi}\,\delta(x-v)\, dv = 2c\sqrt{\pi}\, f(x)$$

Initial condition! $u(x,0) = 2c\sqrt{\pi}\, f(x)$

Summary

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$$

$$u(x,0) = f(x)$$

solution is convolution:

$$u(x,t) = \int_{-\infty}^{\infty} f(v) \underbrace{\frac{1}{2c\sqrt{\pi t}} e^{-\frac{(x-v)^2}{4c^2 t}}}_{\text{Heat kernel}} dv \qquad (*)$$

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$$ 1D heat equation
diffusion

Solution on infinite line with $u(x,0) = f(x)$:

Example: $f(x) = e^{-|x|}$

① $$u(x,t) = \int_0^\infty [A(p)\cos px + B(p)\sin px]\, e^{-c^2p^2t}\, dp$$

$$A(p) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(v)\cos pv\, dv \qquad B(p) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(v)\sin pv\, dv$$

$$A(p) = \frac{1}{\pi}\int_{-\infty}^{\infty} e^{-|v|}\cos(pv)\, dv = \frac{2}{\pi}\int_0^\infty e^{-v}\,\frac{e^{ipv}+e^{-ipv}}{2}\, dv$$

$$= \frac{1}{\pi}\left(\frac{-1}{ip-1} + \frac{-1}{-ip-1}\right) = \frac{1}{\pi}\,\frac{1+ip+1-ip}{(1-ip)(1+ip)} = \frac{2}{\pi}\,\frac{1}{1+p^2}$$

$$B(p) = \frac{1}{\pi}\int_{-\infty}^{\infty} \underbrace{e^{-|v|}}_{\text{even}}\, \underbrace{\sin pv}_{\text{odd}}\, dv = 0$$

$$u(x,t) = \int_0^\infty dp\, \frac{2}{\pi}\,\frac{1}{p^2+1}\cos(px)\, e^{-c^2p^2t}$$

2)

$$u(x,t) = \frac{1}{2c\sqrt{\pi t}} \int_{-\infty}^{\infty} f(v)\, e^{-\frac{(x-v)^2}{4c^2t}}\, dv$$

$$u(x,t) = \frac{1}{2c\sqrt{\pi t}} \int_{-\infty}^{\infty} e^{-|v|}\, e^{-\frac{(x-v)^2}{4c^2t}}\, dv =$$

$$2c\sqrt{\pi t}\, u(x,t) = \int_{-\infty}^{0} e^{v}\, e^{-(x-v)^2/4c^2t}\, dv + \int_{0}^{\infty} e^{-v}\, e^{-(x-v)^2/4c^2t}\, dv$$

$\downarrow u = -v$

$$= \int_{0}^{\infty} e^{-u}\, e^{-(x+u)^2/4c^2t}\, du + \int_{0}^{\infty} e^{-v}\, e^{-(x-v)^2/4c^2t}\, dv$$

$\tilde{u} = \frac{u+x}{2c\sqrt{t}}$

$$= 2c\sqrt{t} \int_{\frac{x}{2c\sqrt{t}}}^{\infty} e^{-2c\sqrt{t}\tilde{u} + x - \tilde{u}^2}\, d\tilde{u} + 2c\sqrt{t} \int_{-\frac{x}{2c\sqrt{t}}}^{\infty} e^{-x - 2c\sqrt{t}\tilde{v} - \tilde{v}^2}\, d\tilde{v}$$

$\tilde{v} = \frac{v-x}{2c\sqrt{t}}$

$$= 2c\sqrt{t} \int_{\frac{x}{2c\sqrt{t}}}^{\infty} e^{-(\underbrace{\tilde{u} + c\sqrt{t}}_{\tilde{\tilde{u}}})^2 + c^2t + x}\, d\tilde{u} + 2c\sqrt{t} \int_{-\frac{x}{2c\sqrt{t}}}^{\infty} e^{-(\underbrace{\tilde{v} + c\sqrt{t}}_{\tilde{\tilde{v}}})^2 + c^2t + x}\, d\tilde{v}$$

$$= 2c\sqrt{t}\, e^{c^2t+x} \frac{\sqrt{\pi}}{2}\left(1 - \operatorname{erf}\left(\frac{x}{2c\sqrt{t}} + c\sqrt{t}\right)\right) + 2c\sqrt{t}\, e^{c^2t-x} \frac{\sqrt{\pi}}{2}\left(1 - \operatorname{erf}\left(c\sqrt{t} - \frac{x}{2c\sqrt{t}}\right)\right)$$

③

$$u(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(\omega)\, e^{-c^2\omega^2 t}\, e^{i\omega x}\, d\omega$$

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(v)\, e^{i v \omega}\, dv$$

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-|v|} e^{i v \omega}\, dv = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{0} e^{v} e^{+iv\omega}\, dv + \frac{1}{\sqrt{2\pi}} \int_{0}^{\infty} e^{-v} e^{iv\omega}\, dv$$

$$= \frac{1}{\sqrt{2\pi}} \int_{0}^{\infty} e^{-v} e^{-iv\omega}\, dv + \frac{1}{\sqrt{2\pi}} \int_{0}^{\infty} e^{-v} e^{iv\omega}\, dv$$

$$= \frac{1}{\sqrt{2\pi}} \frac{-1}{-1-i\omega} + \frac{1}{\sqrt{2\pi}} \frac{-1}{-1+i\omega} = \frac{1}{\sqrt{2\pi}} \left(\frac{1}{1+i\omega} + \frac{1}{1-i\omega} \right)$$

$$= \frac{1}{\sqrt{2\pi}} \frac{1-i\omega + 1+i\omega}{1+\omega^2} = \sqrt{\frac{2}{\pi}} \frac{1}{1+\omega^2}$$

$$u(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega \sqrt{\frac{2}{\pi}} \frac{1}{1+\omega^2} e^{-c^2\omega^2 t} e^{i\omega x}$$

$$= \frac{1}{\pi} \int_{-\infty}^{\infty} d\omega \frac{1}{1+\omega^2} e^{-\omega^2 c^2 t} e^{i\omega x}$$

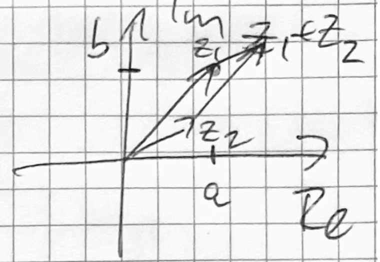
Same as ② ! $\omega \to p$

Simple complex functions

- Complex arythmetic

$$z = \underbrace{a}_{\mathrm{Re}\,z} + i\underbrace{b}_{\mathrm{Im}\,z} \qquad a, b \in \mathbb{R} \quad , \quad i^2 = -1$$

$$z_1 + z_2 = (a_1 + a_2) + i(b_1 + b_2)$$

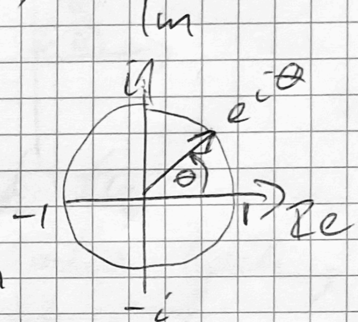


$$z_1 \cdot z_2 = (a_1 + ib_1)(a_2 + ib_2) =$$

$$= (a_1 a_2 - b_1 b_2) + i(a_1 b_2 + a_2 b_1)$$

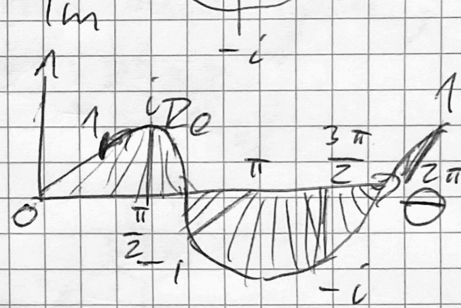
$$|z|^2 = z \cdot \bar{z} = a^2 + b^2 \qquad \frac{1}{i} = -i$$

- Complex exponential



$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$\theta \in \mathbb{R}$$



General argument e^z

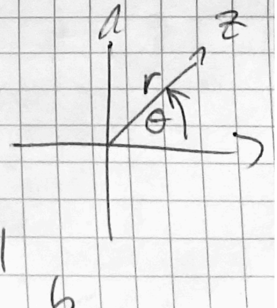
$$e^z = e^{a+ib} = e^a \cdot e^{ib} =$$

$$= \underbrace{e^a}_{} \cdot \underbrace{(\cos b + i\sin b)}_{}$$

| real part | imag part | of z |
| magnitude | phase | of e^z |

periodic in $\mathrm{Im}\,z$

$$e^z = e^{z + 2\pi i} = e^z \cdot \underbrace{e^{2\pi i}}_{1}$$



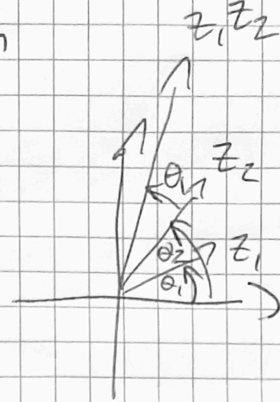
- Polar form $\quad z = r \cdot e^{i\theta} \qquad r = |z|$

principal argument $\quad -\pi < \theta \leq \pi \qquad \tan\theta = \frac{b}{a}$

Multiplication in polar form

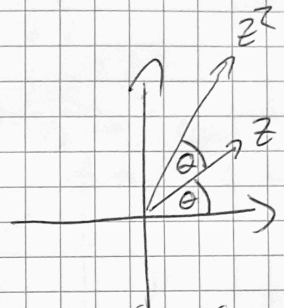
$$z_1 z_2 = r_1 r_2 \, e^{i(\theta_1+\theta_2)}$$

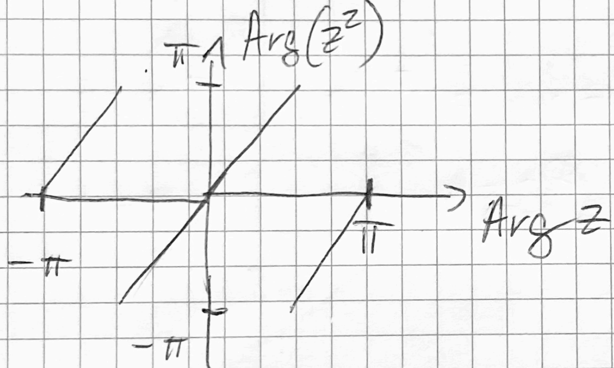
argument added,
magnitude multiplied



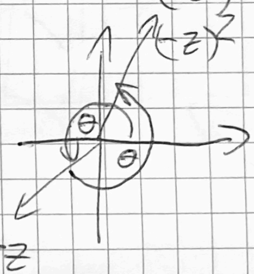
- Example $f(z) = z^2$

$$z = r\,e^{i\theta}$$
$$z^2 = r^2\,e^{2i\theta}$$





Winds around twice
takes every value twice,
because $z^2 = (-z)^2$



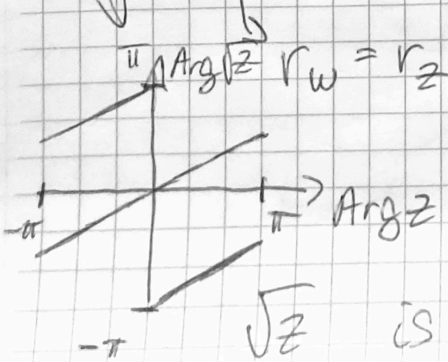
rotate graph

- Example $f(z) = \sqrt{z}$

What is $\sqrt{\ }$?

def: $w = \sqrt{z}$ if $w^2 = z$

(but we saw that w and $-w$ both work!

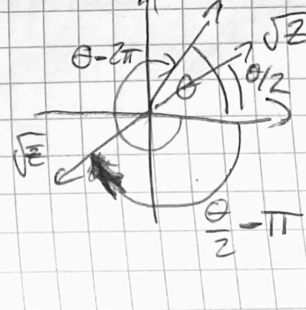


$r_w = r_z$, $2\theta_w = \theta_z \mod 2\pi$

$$\theta_w = \frac{\theta_z}{2} \mod \pi$$

$$\theta_w = \frac{\theta_z}{2} \text{ or } \frac{\theta_z}{2} \pm \pi$$

$\sqrt{z}$ is double valued



- Example complex logarithm

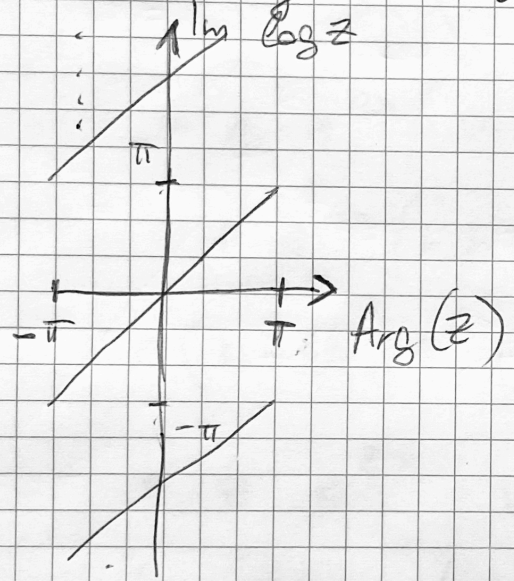
$\log z = w$ if $\exp w = z$

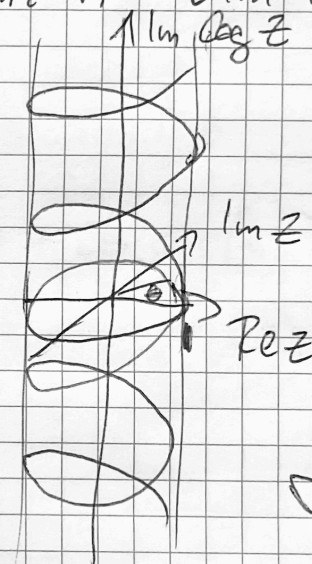
If true for w, true for $w + 2\pi i \cdot n$ $\quad n \in \mathbb{Z}$

infinite-valued!

Let $\quad z = e^{i\theta} = \mathrm{Arg}(z)$

$$\log z = i\theta + 2\pi i \cdot n = i \,\mathrm{Im} \log z$$

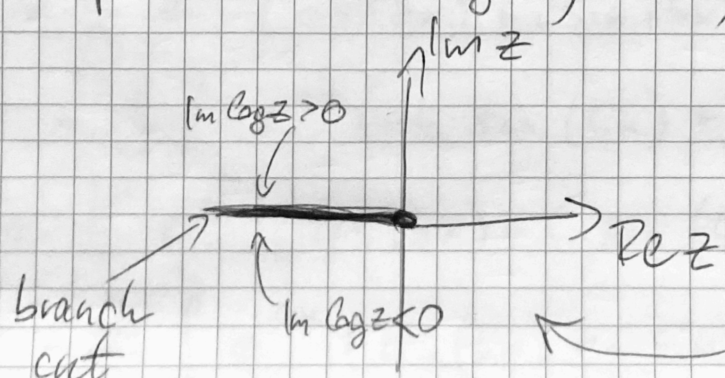




Make it unique: $\quad -\pi < \mathrm{Im} \log z \leq \pi$

principal log

Jumps where $\mathrm{Arg}(z) = \pm\pi$, negative real line



"spiral staircase" on floorplan

3.3/18

$$f(z) = \frac{z-i}{z+i}$$

$$f(i) = 0$$

$$f(i+\Delta z) = \frac{\Delta z}{2i+\Delta z} = \frac{\Delta z}{2i} + O(\Delta z^2) \longrightarrow f'(i) = \frac{1}{2i}$$

More generally (?)

$$f(z+\Delta z) = \frac{z+\Delta z - i}{z+\Delta z+i} = \frac{z-i+\Delta z}{(z+i)\left(1+\frac{\Delta z}{z+i}\right)} = \frac{z-i+\Delta z}{z+i}\left(1 - \frac{\Delta z}{z+i} + O(\Delta z^2)\right)$$

$$= \underbrace{\frac{z-i}{z+i}}_{f(z)} + \frac{\Delta z}{z+i} - \frac{z-i}{(z+i)^2}\Delta z + O(\Delta z^2)$$

$$\lim_{\Delta z\to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z} = \frac{1}{z+i} - \frac{z-i}{(z+i)^2} = \frac{2i}{(z+i)^2}$$

agrees with usual diff. rules

$$f'(z=i) = \frac{2i}{(i+i)^2} = \frac{1}{2i} = -\frac{i}{2}$$

3.3/21 $\quad f(z) = i(1-z)^n \qquad f'(0) = ?$

$$f(0) = i,$$

$$f(0+\Delta z) = i(1-\Delta z)^n = i(1-\Delta z)\dots(1-\Delta z)$$

$$= i(1 - n\cdot\Delta z + O(\Delta z^2))$$

$$\to f'(0) = \frac{i(\cancel{1}-n\Delta z)\cancel{-i}}{\Delta z} = -in$$

using diff. rules:

$$f'(z) = i\cdot n(1-z)^{n-1}\cdot(-1) \quad \to f'(0) = -i\cdot n \quad ✓$$

Analytic functions, Cauchy-Riemann equations

$$f(z) = u(z) + i\,v(z) \qquad u, v \in \mathbb{R}$$

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z} = \begin{cases} u_x + i v_x & \Delta y = 0 \\ -i u_y + v_y & \Delta x = 0 \end{cases}$$

$$\rightarrow \quad u_x = v_y \qquad u_y = -v_x$$

But then

$$f'(z) = u_x - i u_y = v_y + i v_x$$

$$u_{xx} = v_{xy} \qquad u_{yy} = -v_{xy}$$

$$\rightarrow \quad u_{xx} + u_{yy} = \Delta u = 0$$

$$u_{xy} = v_{yy} \qquad u_{yx} = -v_{xx}$$

$$v_{xx} + v_{yy} = \Delta v = 0$$

u, v are harmonic functions

Is $f(z) = \cos(z)$ analytic (holomorphic)?

$$\cos z = \frac{e^{iz} + e^{-iz}}{2} \qquad \sin z = \frac{e^{iz} - e^{-iz}}{2i}$$

$$e^{iz} = \cos z + i \sin z$$

$f(z) = u(z) + i\,v(z)$? $\quad u, v \in \mathbb{R}$

$$\cos z = \frac{1}{2}\left(e^{i(x+iy)} + e^{-i(x+iy)}\right) =$$

$$= \frac{1}{2} e^{-y}(\cos x + i \sin x) + \frac{1}{2} e^{y}(\cos x - i \sin x)$$

$$= \cos x \cdot \frac{e^{y} + e^{-y}}{2} - i \sin x \cdot \frac{e^{y} - e^{-y}}{2}$$

$$= \underbrace{\cos x \cdot \cosh y}_{u(z)} - i \cdot \underbrace{\sin x \cdot \sinh y}_{-v(z)}$$

Cauchy – Riemann:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \qquad\qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$-\sin x \cdot \cosh y \overset{\checkmark}{=} -\sin x \cdot \cosh y \qquad\qquad \cos x \sinh y \overset{\checkmark}{=} \cos x \cdot \sinh y$$

——— o ———

Is $f(z) = \mathrm{Re}(z^2) - i\,\mathrm{Im}(z^2)$ analytic? 13.4/5

$$(x+iy)^2 = x^2 - y^2 + 2ixy$$

$$f(z) = \underbrace{x^2 - y^2}_{u} - i \underbrace{2xy}_{-v} = \bar{z}^2 = \overline{z^2}$$

$$\partial_x u = 2x \neq \partial_y v = -2x$$

$$\partial_y u = -2y \neq -\partial_x v = 2y$$

it is NOT holomorphic!

(involves conjugation)

Harmonic function: solution of Laplace's equation

has continuous 2^{nd} partial derivatives

$v = x \cdot y$ 13.4/14

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0 + 0 = 0 \quad \checkmark \qquad \text{harmonic}$$

Find u: $\frac{\partial u}{\partial x} = x$, $\frac{\partial u}{\partial y} = -y$

$u = \frac{x^2}{2} + C(y) \rightarrow C'(y) = -y$

$C(y) = -y^2/2 + \tilde{C}$

$$u = \frac{x^2 - y^2}{2} + \tilde{C}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 1 - 1 = 0 \quad \checkmark \qquad \text{harmonic}$$

$$\rightarrow \quad f(z = x + iy) = \frac{x^2 - y^2}{2} + ixy + \tilde{C} = \frac{1}{2}(x + iy)^2 + \tilde{C} = \frac{z^2}{2} + \tilde{C}$$

is a holomorphic function